BIOS 731: HW4

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Problem 0

GitHub Repo: https://github.com/rndparr/bios731_hw4_parrish

Bayesian K-Mixture Model

We consider a Bayesian Mixture Model of unit-variance univariate Gaussians. There are:

- K mixture components, corresponding to K Gaussian distributions with means $\boldsymbol{\mu} = \{\mu_1, \ldots, \mu_K\}$
- cluster assignments $c_i \in \{1, \ldots, n\}$
- observations $y_i \in \{1, \ldots, n\}$

Data generating model is

$$
\begin{aligned}
\mu &\sim N(0, \sigma^2) \\
c_i &\sim \text{Categorical}(1/K. \ldots, 1/K) \\
y_i | c_i, \boldsymbol{\mu} &\sim N(\mathbf{c}_i^T \boldsymbol{\mu}, 1), \qquad\qquad \mathbf{c}_i = [c_{ik}]_{K\times 1} = [I(c_i = k)]_{K\times 1}
\end{aligned}
$$

Problem 1: Gibbs sampler derivations and implementation

Posterior distribution of cluster assignments

$$
\begin{aligned}
p(c_i = k | y_i, \boldsymbol{\mu}) &= \frac{p(y_i | c_i = k, \boldsymbol{\mu})}{p(y_i | \boldsymbol{\mu})}\, p(c_i = k) \\
&= \frac{p(y_i | c_i = k, \boldsymbol{\mu})}{\sum_{j=1}^K p(y_i | c_i = j, \boldsymbol{\mu}) p(c_i = j)}\, p(c_i = k) \\
&= \frac{\exp\left\{-\frac{1}{2}(y_i - \mu_k)^2\right\}}{\sum_{j=1}^K \exp\left\{-\frac{1}{2}(y_i - \mu_j)^2\right\} \frac{1}{K}} \frac{1}{K} \\
&= \frac{\exp\left\{-\frac{1}{2}(y_i - \mu_k)^2\right\}}{\sum_{j=1}^K \exp\left\{-\frac{1}{2}(y_i - \mu_j)^2\right\}} \\
&= \pi_{ik} \\
c_i | y_i, \boldsymbol{\mu} &\sim \text{Categorical}(\pi_{i1}, \ldots, \pi_{iK}) = \text{Categorical}(\boldsymbol{\pi}_i)
\end{aligned}
$$

Posterior distribution of cluster means

First, let n_k be the number of c_i such that $c_i = k$.

$$
\begin{aligned}
p(\mu_k|\mathbf{y},\mathbf{c}) &\propto p(\mu_k) \prod_{i:c_i=k} p(y_i|c_i,\mu_k) \\
&= \exp\left\{-\frac{\mu_k^2}{2\sigma^2}\right\} \prod_{i:c_i=k} \exp\left\{-\frac{1}{2}(y_i-\mu_k)^2\right\} \\
&= \exp\left\{-\frac{\mu_k^2}{2\sigma^2}\right\} \exp\left\{\sum_{i:c_i=k} -\frac{1}{2}(y_i-\mu_k)^2\right\} \\
&= \exp\left\{-\frac{1}{2}\left(\frac{\mu_k^2}{\sigma^2} + \sum_{i:c_i=k}(y_i-\mu_k)^2\right)\right\} \\
&= \exp\left\{-\frac{\frac{1}{\sigma^2}+n_k}{2}\left(\mu_k - \frac{1}{\frac{1}{\sigma^2}+n_k}\sum_{i:c_i=k} y_i\right)^2\right\}
\end{aligned}
$$

$$
\mu_k|\mathbf{y},\mathbf{c} \sim N\big(\mu_k^*, \sigma_k^{*2}\big) \qquad \sigma_k^{*2} = \left(\frac{1}{\sigma^2}+n_k\right)^{-1} \qquad \mu_k^* = \sigma_k^{*2}\sum_{i:c_i=k} y_i
$$

Problem 2: Variational inference derivations

Mean-field variational approximation to the posterior distribution:

Joint probability

$$
\begin{aligned}
p(\boldsymbol{\mu},\mathbf{c},\mathbf{y}) &= p(\boldsymbol{\mu})\, p(\mathbf{c},\mathbf{y}) = p(\boldsymbol{\mu}) \prod_{i=1}^{n} p(c_i)\, p(y_i|c_i,\boldsymbol{\mu}) \\
\log p(\boldsymbol{\mu},\mathbf{c},\mathbf{y}) &= \sum_{k=1}^{K} \log p(\mu_k) + \sum_{i=1}^{n} \log p(c_i) + \sum_{i=1}^{n} p(y_i|c_i,\boldsymbol{\mu}) \\
&= \sum_{k=1}^{K}\left\{-\frac{1}{2}\log(2\pi\sigma^2) - \frac{\mu_k^2}{2\sigma^2}\right\} - \sum_{i=1}^{n}\log(K) + \sum_{i=1}^{n}\left\{-\frac{1}{2}\log(2\pi) - \frac{(y_i-\mu_k)^2}{2}\right\}
\end{aligned}
$$

Mean-field approximation

$$
\begin{aligned}
q(\boldsymbol{\mu},\mathbf{c}) &= \prod_{k=1}^{K} q(\mu_k;\mu_k^*,\sigma_k^{*2}) \prod_{i=1}^{n} q(c_i;\boldsymbol{\pi}_i) \\
q(\mu_k;\mu_k^*,\sigma_k^{*2}) &= N\big(\mu_k^*,\sigma_k^{*2}\big) \\
q(c_i;\boldsymbol{\pi}_i) &= \text{Categorical}(\boldsymbol{\pi}_i), \qquad \sum_{k=1}^{K}\pi_{ik} = 1
\end{aligned}
$$

Update for cluster assignment

$$
\begin{aligned}
q^*(c_i = k; \boldsymbol{\pi}_i) &\propto \exp\left\{\log p(c_i) + \mathrm{E}\left[\log p(y_i|c_i, \boldsymbol{\mu});\ \boldsymbol{\mu}^*, \boldsymbol{\sigma}^{*2}\right]\right\} \\
&= \exp\left\{\mathrm{E}\left[\log p(y_i|\mu_k);\ \mu_k^*, \sigma_k^{*2}\right]\right\} + \text{const.} \\
&= \exp\left\{\mathrm{E}\left[-\frac{(y_i - \mu_k)^2}{2};\ \mu_k^*, \sigma_k^{*2}\right]\right\} + \text{const.} \\
&= \exp\left\{\mathrm{E}\left[-\frac{1}{2}y_i^2 \mu_k y_i - \frac{1}{2}\mu_k^2;\ \mu_k^*, \sigma_k^{*2}\right]\right\} + \text{const.} \\
&= \exp\left\{\mathrm{E}\left[\mu_k y_i - \frac{1}{2}\mu_k^2;\ \mu_k^*, \sigma_k^{*2}\right]\right\} + \text{const.} && \text{(removed } y_i^2 \text{ term, const. wrt } c_i\text{)} \\
&= \exp\left\{y_i \mathrm{E}\left[\mu_k;\ \mu_k^*, \sigma_k^{*2}\right] - \frac{1}{2}\mathrm{E}\left[\mu_k^2;\ \mu_k^*, \sigma_k^{*2}\right]\right\} + \text{const.} \\
&= \exp\left\{y_i\, \mu_k^* - \frac{1}{2}\left(\sigma_k^{*2} + \mu_k^{*2}\right)\right\} + \text{const.}
\end{aligned}
$$

$$
\pi_{ik} = \frac{\exp\left\{y_i \mu_k^* - \frac{1}{2}\left(\sigma_k^{*2} + \mu_k^{*2}\right)\right\}}{\sum_{j=1}^K \exp\left\{y_i \mu_j^* - \frac{1}{2}\left(\sigma_j^{*2} + \mu_j^{*2}\right)\right\}} \qquad \text{(normalized)}
$$

Update for cluster means

$$
\begin{aligned}
q(\mu_k; \mu_k^*, \sigma_k^{*2}) &\propto \exp\left\{\log p(\mu_k) \sum_{i=1}^n \mathrm{E}\left[\log p(y_i|c_i, \boldsymbol{\mu});\ \boldsymbol{\pi}_i, \boldsymbol{\mu}_{-k}^*, \boldsymbol{\sigma}_{-k}^{*2}\right]\right\} \\
&= \exp\left\{\mu_k \left(\sum_{i=1}^n \pi_{ik} y_i\right) - \mu_k^2 \left(\frac{\sigma^2}{2} + \frac{1}{2}\sum_{i=1}^n \pi_{ik}\right)\right\} + \text{const.} \\
\log q(\mu_k; \mu_k^*, \sigma_k^{*2}) &= \mu_k \left(\sum_{i=1}^n \pi_{ik} y_i\right) - \mu_k^2 \left(\frac{\sigma^2}{2} + \frac{1}{2}\sum_{i=1}^n \pi_{ik}\right) + \text{const.} \\
0 &= \frac{\partial \log q(\mu_k; \mu_k^*, \sigma_k^{*2})}{\partial \mu_k} \\
&= \sum_{i=1}^n \pi_{ik} y_i - 2\mu_k \left(\frac{\sigma^2}{2} + \frac{1}{2}\sum_{i=1}^n \pi_{ik}\right) + \text{const.} \\
&= \sum_{i=1}^n \pi_{ik} y_i - \mu_k \left(\sigma^2 + \sum_{i=1}^n \pi_{ik}\right) + \text{const.}
\end{aligned}
$$

$$
\begin{aligned}
\mu_k^* = \hat{\mu}_k &= \frac{\sum_{i=1}^n \pi_{ik} y_i}{\sigma^{-2} + \sum_{i=1}^n \pi_{ik}} \\
\sigma_k^{*2} = \hat{\sigma}_k^2 &= \frac{1}{\sigma^{-2} + \sum_{i=1}^n \pi_{ik}}
\end{aligned}
$$

Problem 3: Fit to data

```r
# load scripts
source(here('source', 'utility.R'))
source(here('source', 'gibbs_functions.R'))
source(here('source', 'cavi_functions.R'))

# import data
```

```r
data(faithful)
y <- faithful[['waiting']]

# run Gibbs Sampler
n_chains <- 4
faithful_gibbs_time <- func_time(
    gibbs_res <- gibbs_multi_chain(
        y, K = 2, sigma2 = 1,
        n_chains = n_chains, n_iter = 5000, burnin = 1000)
    )

# run CAVI
faithful_cavi_time <- func_time(
    cavi_res <- cavi_model(
        y, K = 2, sigma2 = 1,
        mu_star = NULL, sigma2_star = NULL,
        tol = 1e-5, max_iter = 1000)
)
```

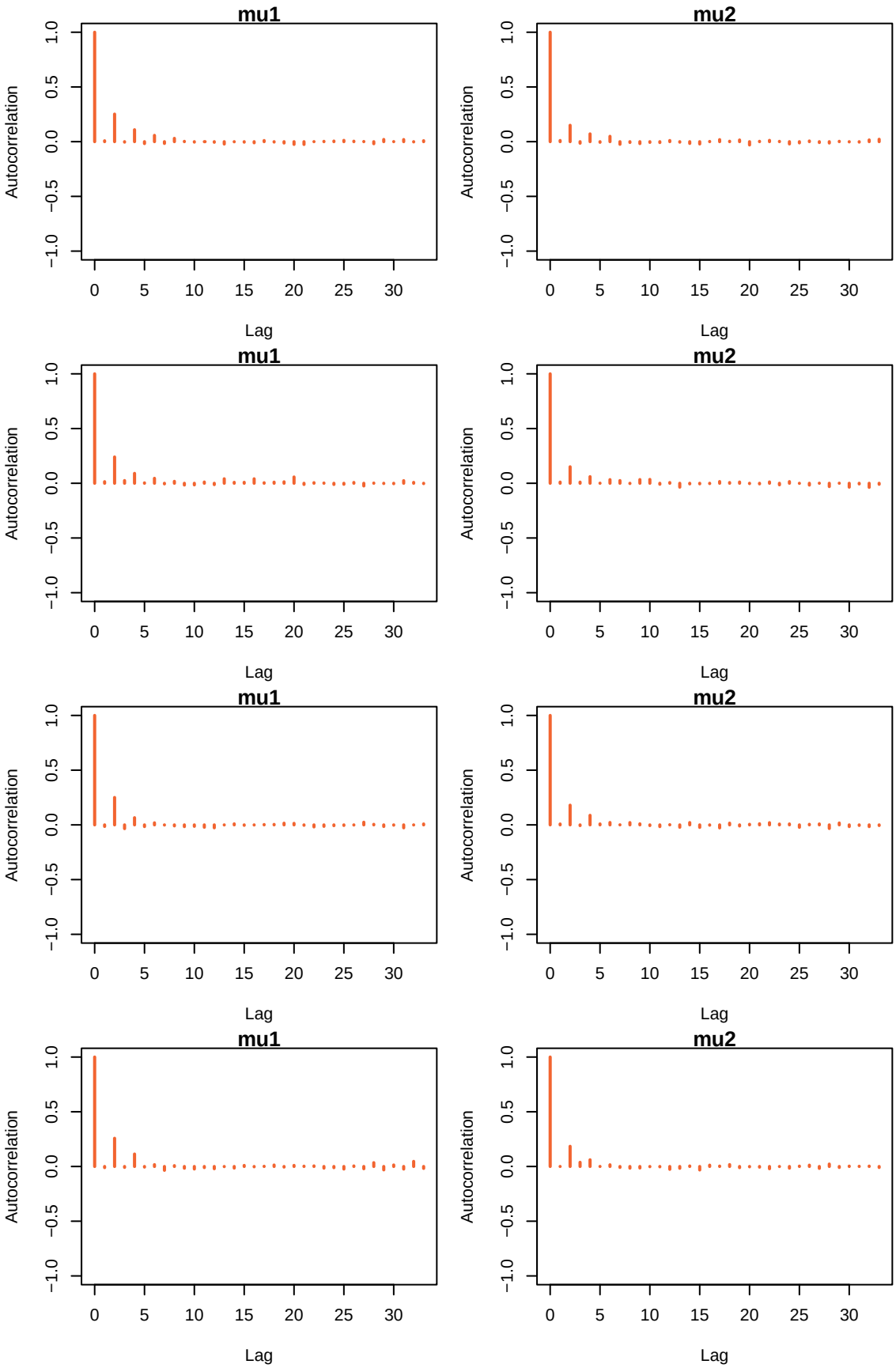

Figure 1: Autocorrelation Plots for Gibbs Sampler; each row represents one chain

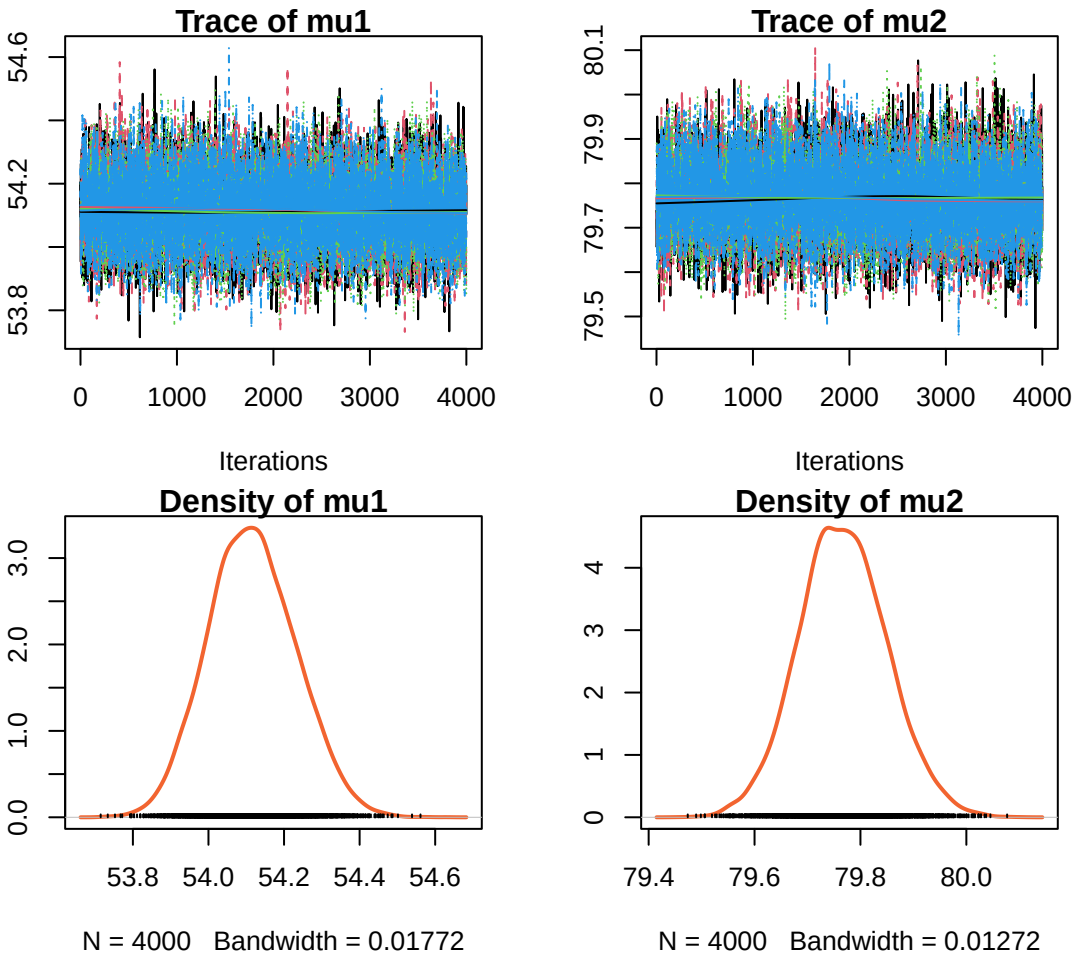

Figure 2: Trace, Density Plots for Gibbs Sampler

Table 1: Effective Sample Size for Gibbs

μ_1	μ_2
8967.323	10163.68

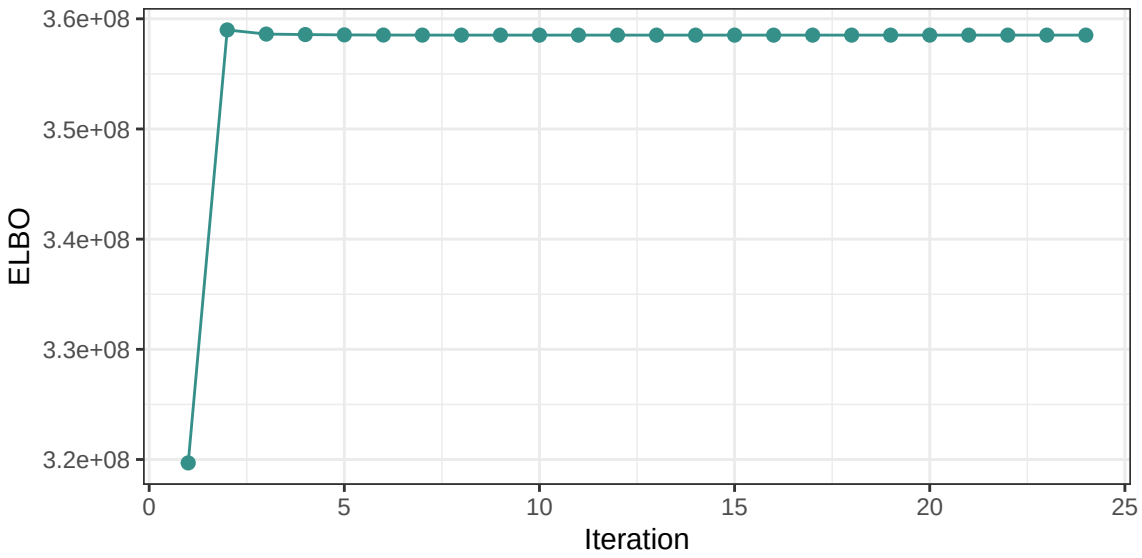

Figure 3: CAVI Convergence

Table 2: Computation Time (Seconds) for Old Faithful Waiting Time Data

CAVI	Gibbs (total)	chain 1	chain 2	chain 3	chain 4
0.023	19.304	4.845	4.712	4.743	4.849

Table 3: Final Estimates of $\boldsymbol{\mu}$

	$\hat{\mu}_1$	$\hat{\mu}_2$
CAVI	54.09716	79.75688
Gibbs	54.11589	79.76743

CAVI was substantially faster than Gibbs Sampling; CAVI took a fraction of a second whereas each Gibbs chain took around 5 seconds. CAVI converged very quickly for this data. Both yielded similar final estimates. The effective sample size for the Gibbs Sampler running for 10000 iterations and 4 chains was around 9.5k to 10.5k.

Problem 4: Perform a simulation study on the cluster

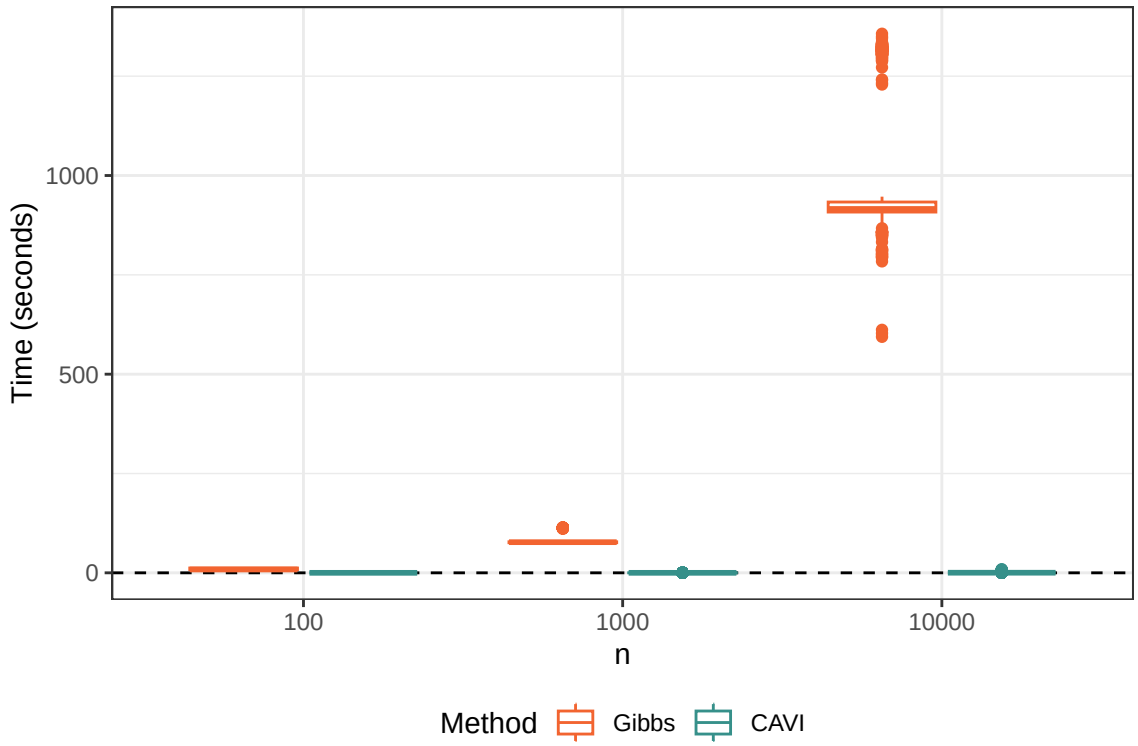

Figure 4: Computation time for simulation.

Table 4: Computation Time (seconds) (Mean ± SD)

n	Gibbs	CAVI
100	10.167 ± 2.032	0.039 ± 0.008
1000	79.887 ± 10.138	0.054 ± 0.015
10000	994.418 ± 173.296	0.492 ± 1.075

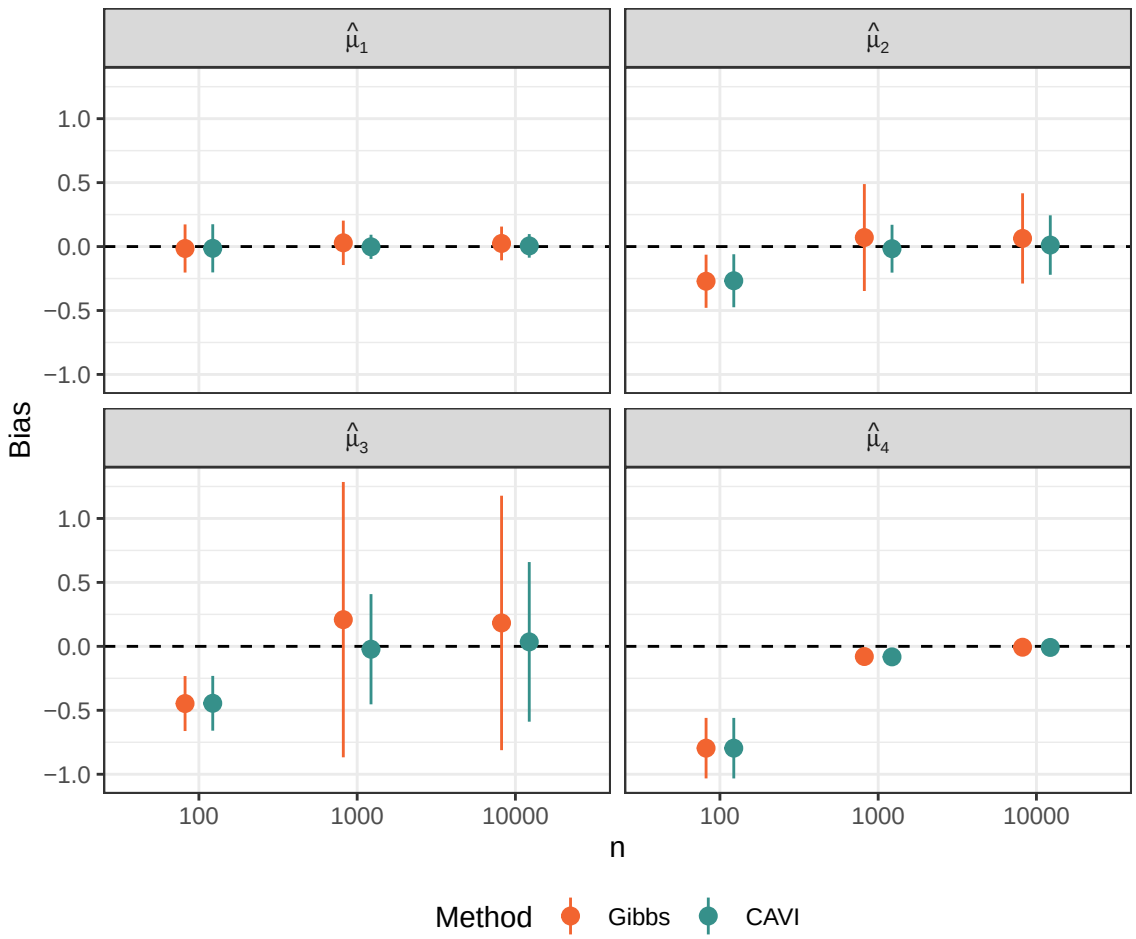

Figure 5: Bias for each $\hat{\mu}_k$.

Table 5: Bias (Mean ± SD)

n	μ_k	Gibbs	CAVI
100	1	-0.015 ± 0.188	-0.014 ± 0.188
	2	-0.271 ± 0.208	-0.267 ± 0.207
	3	-0.447 ± 0.215	-0.445 ± 0.214
	4	-0.796 ± 0.237	-0.796 ± 0.237
1000	1	0.029 ± 0.174	-0.002 ± 0.095
	2	0.071 ± 0.418	-0.017 ± 0.187
	3	0.209 ± 1.077	-0.022 ± 0.431
	4	-0.08 ± 0.065	-0.082 ± 0.065
10000	1	0.024 ± 0.132	0.005 ± 0.092
	2	0.064 ± 0.353	0.012 ± 0.232
	3	0.183 ± 0.995	0.035 ± 0.624
	4	-0.007 ± 0.024	-0.008 ± 0.023

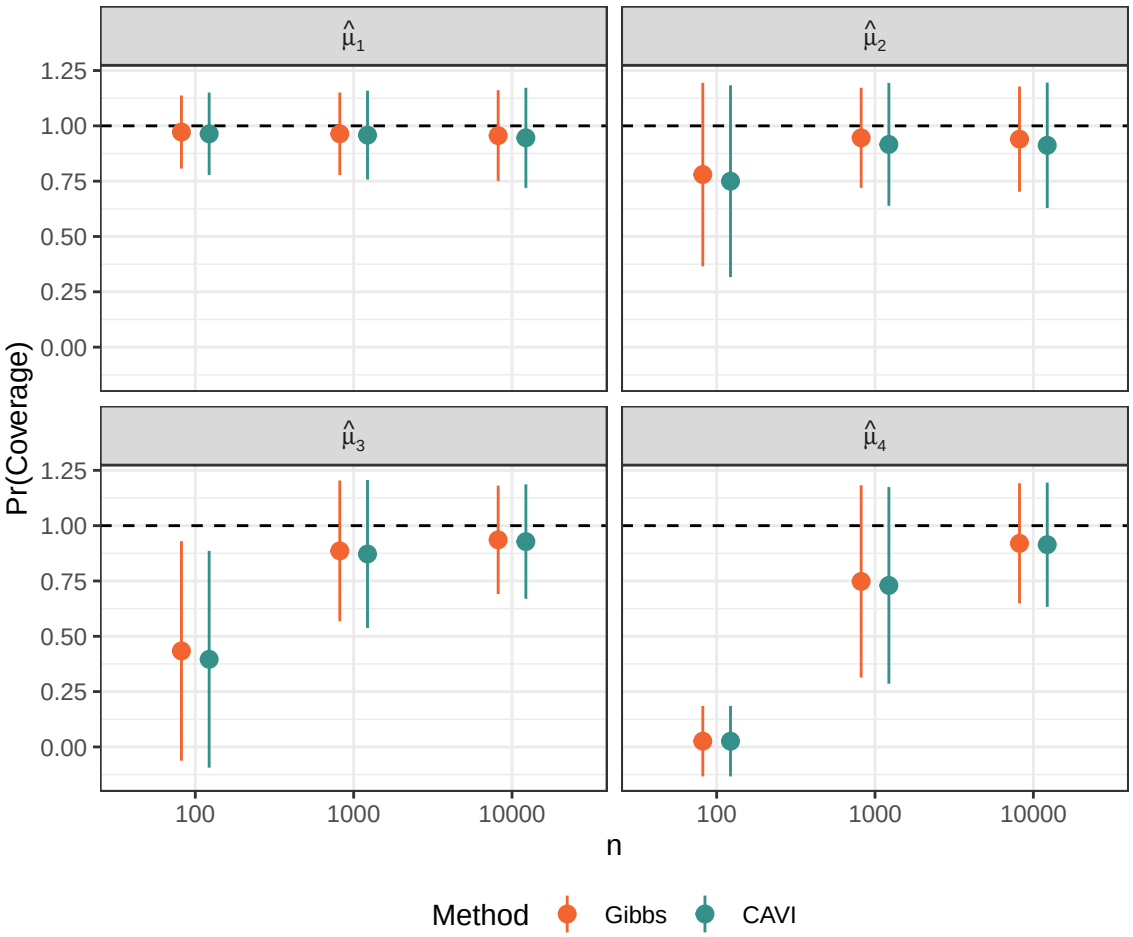

Figure 6: Coverage for each for each $\hat{\mu}_k$.

Table 6: Coverage (Mean ± SD)

n	μ_k	Gibbs	CAVI
100	1	0.972 ± 0.165	0.964 ± 0.186
	2	0.78 ± 0.415	0.75 ± 0.433
	3	0.434 ± 0.496	0.396 ± 0.49
	4	0.026 ± 0.159	0.026 ± 0.159
1000	1	0.964 ± 0.186	0.958 ± 0.201
	2	0.946 ± 0.226	0.916 ± 0.278
	3	0.886 ± 0.318	0.872 ± 0.334
	4	0.748 ± 0.435	0.73 ± 0.444
10000	1	0.956 ± 0.205	0.946 ± 0.226
	2	0.94 ± 0.238	0.912 ± 0.284
	3	0.936 ± 0.245	0.928 ± 0.259
	4	0.92 ± 0.272	0.914 ± 0.281

CAVI was once again substantially faster than even single-chain Gibbs Sampling; CAVI took a fraction of a second for $n = 100, 1000$ and less than 2 seconds for $n = 10000$ whereas, on average, Gibbs sampling took about 10 seconds for $n = 100$, 1.5 minutes for $n = 1000$ and around 17 minutes for $n = 10000$.

Performance for μ_1 was comparable for all sample sizes. However, for $n = 100$, both methods underestimated the other μ_k values; The sample size of $n = 100$ was not sufficient for either method to accurately estimate μ_3 and μ_4; the proportion of coverage for was less than 50% for μ_3 and near 0% for μ_4 for this sample size. Gibbs sampling overestimated μ_2 and μ_3 for greater sample sizes while CAVI seemed more accurate. Both CAVI and Gibbs had very high standard deviation for μ_3 for these larger sample sizes.